

# A Silicon Photonics Mach-Zehnder Interferometer

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## I. INTRODUCTION

Silicon photonics is a rapidly growing industry that has already seen large-scale commercialization in the telecommunications industry. Emerging and promising applications include biomedical sensing and quantum information processing. This course aims to teach the fundamentals of the design, fabrication, and testing of silicon photonic devices. The coursework takes a rapid approach intended to quickly introduce the student to the entire process including design, simulations, layout, layout review, fabrication, and automated testing and data analysis. In order to proceed quickly but still give meaningful hands-on experience, the course centers around a simple silicon photonic device: the Mach-Zehnder interferometer.

In a Mach-Zehnder interferometer, light is split and travels down two paths and then is recombined. If the two waveguides are not exactly identical in their lengths  $L$  or propagation constants  $\beta$  then the light will be in or out of phase when it is recombined. If the light from the two paths is in-phase there will be constructive interference and the light intensity in the combined path will be the same as that in the initial path. If the light is perfectly out of phase when recombined, destructive interference will occur. In a silicon photonic waveguide, the result of destructive interference is that the light couples into higher-order, dispersive waveguide modes and is lost so that the output waveguide has no light. The output of the interferometer is typically measured over a wavelength sweep so that oscillations between constructive and destructive interference are clearly visible. The period of the oscillations is known as the free spectral range (FSR) and is a key parameter for characterizing the interferometer.

In this report, two waveguide parameters are varied in order to create a mismatch between the two waveguide arms of the interferometer, namely (1) the lengths and (2) the widths of the two waveguides. Varying the lengths creates a path-length mismatch  $\Delta L$ , while varying the widths creates a mismatch in the propagation constant  $\beta$  because the waveguide width changes the effective index. These effects and their relationship to the FSR are described in Sec. II.

The interferometer FSR is directly connected to the group index  $n_g$  of the waveguide. In the analysis of experimental data in Sec. ??,  $n_g$  will be determined based on the measurements and compared with simulation results presented in Sec. III. This will provide insight into the effects of manufacturing variability and the accuracy of simulations. These are important things to take into consideration in the development of all silicon photonic devices.

## II. THEORY

This section begins with a description of a compact polynomial model for a waveguide index of refraction. It then

details the mathematical form of an interferometer transfer function and free spectral range, including the dependence of the FSR on group index  $n_g$ . The equations for the FSR are initially taken from the book "Silicon Photonics Design" by Chrostowski and Hochberg [1] which only describes interferometer path mismatch based on differences in waveguide length. Because this report will also explore interferometers where the two paths have identical lengths but different widths, a more detailed derivation is presented for the FSR of this type of interferometer. Following this derivation, tables are given which outline the parameter variations and predicted FSR for all devices which will be tested.

### A. Compact waveguide model

The properties of a waveguide can be described using a compact polynomial model for the index of refraction, as described in [1]. The index of refraction  $n$  is Taylor expanded as a function of wavelength  $\lambda$  ( $\mu\text{m}$ ) as

$$n(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2. \quad (1)$$

The parameters  $n_1$  (unit-less),  $n_2$  ( $\mu\text{m}^{-1}$ ), and  $n_3$  ( $\mu\text{m}^{-2}$ ) can be obtained through polynomial fitting (such as a least-squares fit method) once  $n(\lambda)$  is calculated through a finite-difference eigenmode (FDE) or similar simulation of a waveguide.

As described in [1], these parameters can also be converted to the more traditional parameters

$$\begin{aligned} n_{\text{eff}} &= n_1 \\ n_g &= n_1 - n_2 \cdot \lambda_0 \\ D &= -2 \cdot \lambda_0 \cdot n_3 / c \end{aligned} \quad (2)$$

where  $n_{\text{eff}}$  is the waveguide effective index,  $n_g$  is the group index, and  $D$  is the waveguide dispersion.

### B. Interferometer transfer function and free spectral range

The transfer function for a simple Mach-Zehnder interferometer is given by equation (4.19c) in [1] as

$$\frac{I_o}{I_i} = \frac{1}{2} [1 + \cos(\beta_1 L_1 - \beta_2 L_2)] \quad (3)$$

$$\beta = \frac{2\pi n(\lambda)}{\lambda} \quad (4)$$

Here  $\beta_1$  and  $\beta_2$  are the wavenumbers of the two waveguides and  $L_1$  and  $L_2$  are the lengths of the waveguides. This equation neglects overall loss and applies to single-mode waveguides where only the total field intensity is considered in each waveguide.

If the imbalance in the two interferometer arms is created by a waveguide length mismatch  $\Delta L = L_2 - L_1$ , then the

free spectral range (FSR) will be given is given (in meters) by equation (4.20b) in [1] as

$$\text{FSR} = \frac{\lambda_0^2}{n_g \Delta L} \quad (5)$$

where  $\lambda_0$  is the wavelength around which the FSR is measured.

In Sec. ?? the group index of the fabricated waveguides will be determined based on the FSR found in the experimental data. This is be easily obtained by solving Eq. (5) for  $n_g$  such that

$$n_g = \frac{\lambda^2}{\text{FSR} \cdot \Delta L}. \quad (6)$$

To extract  $n_g$  from the experimental data, the period of the interferometer fringes (the FSR) will be determined numerically around a chosen wavelength  $\lambda$  (e.g. 1550 nm). The measured FSR, the chosen  $\lambda$ , and the known  $\Delta L$  from the layout will be substituted into Eq. (6) to determine the experimentally-measured waveguide group index.

### C. Free spectral range for a waveguide-width-mismatched interferometer

To determine the FSR for devices which have a mismatch in waveguide width (so that  $n_1 \neq n_2$ ), but no mismatch in the waveguide length ( $L_1 = L_2$ ), we start with Eq. (3) The period (FSR) of the transfer function is determined by the phase

$$\delta = (\beta_1 - \beta_2)L.$$

The FSR is the wavelength difference between adjacent peaks  $\Delta\lambda = \lambda_{m+1} - \lambda_m$ . The phase difference between adjacent peaks is always  $2\pi$ , or in other words,

$$\begin{aligned} 2\pi &= \delta_m - \delta_{m+1} \\ &= (\beta_{1,m} - \beta_{2,m})L - (\beta_{1,m+1} - \beta_{2,m+1})L. \end{aligned}$$

Rearranging and writing

$$\begin{aligned} \Delta\beta_1 &= \beta_{1,m} - \beta_{1,m+1} \\ \Delta\beta_2 &= \beta_{2,m} - \beta_{2,m+1} \end{aligned}$$

we have

$$\frac{2\pi}{L} = \Delta\beta_1 - \Delta\beta_2. \quad (7)$$

We can approximate that  $\Delta\beta$  varies linearly with  $\lambda$  as

$$\Delta\beta = -\frac{d\beta}{d\lambda}\bigg|_{\lambda_0} \Delta\lambda$$

and we can evaluate  $\frac{d\beta}{d\lambda}$  at  $\lambda_0$  since  $\beta = \frac{2\pi n(\lambda)}{\lambda}$ , which is

$$\frac{d\beta}{d\lambda}\bigg|_{\lambda_0} = -\frac{2\pi}{\lambda_0^2} \left( n - \frac{dn}{d\lambda}\bigg|_{\lambda_0} \lambda_0 \right) = -\frac{2\pi}{\lambda_0^2} n_g.$$

Plugging these into Eq. (7) we have

$$\frac{2\pi}{L} = \frac{2\pi}{\lambda_0^2} (n_{1,g} - n_{2,g}) \Delta\lambda. \quad (8)$$

Which we can solve for  $\Delta\lambda$  (the FSR):

$$\text{FSR} = \Delta\lambda = \frac{\lambda_0^2}{L \Delta n_g}. \quad (9)$$

TABLE I  
INTERFEROMETERS WITH LENGTH-MISMATCH

| Device num | $w$    | $\Delta L$        | $n_{\text{eff}}$ | $n_g$ | FSR     |
|------------|--------|-------------------|------------------|-------|---------|
| 1          | 500 nm | 100 $\mu\text{m}$ | 2.45             | 4.20  | 5.72 nm |
| 2          | 500 nm | 200 $\mu\text{m}$ | 2.45             | 4.20  | 2.86 nm |
| 3          | 450 nm | 100 $\mu\text{m}$ | 2.35             | 4.29  | 5.60 nm |
| 4          | 550 nm | 100 $\mu\text{m}$ | 2.52             | 4.12  | 5.83 nm |
| 5          | 600 nm | 100 $\mu\text{m}$ | 2.57             | 4.06  | 5.92 nm |

TABLE II  
INTERFEROMETERS WITH WIDTH-MISMATCH

| Device num | $w_1$  | $w_2$  | $L$               | FSR     |
|------------|--------|--------|-------------------|---------|
| 6          | 500 nm | 450 nm | 900 $\mu\text{m}$ | 27.9 nm |
| 7          | 500 nm | 550 nm | 900 $\mu\text{m}$ | 34.1 nm |
| 8          | 500 nm | 600 nm | 900 $\mu\text{m}$ | 19.0 nm |

### D. Device parameter variations

In order to evaluate the effects of changing various parameters, the Mach-Zehnder interferometers will be designed, simulated, and fabricated across a sweep of waveguide *length* mismatches  $\Delta L$  and a sweep of waveguide *width* mismatches. These parameters are shown in Tables I and II.

The devices in Table I have varying length mismatches between the two waveguides in the interferometer, but the widths of the two waveguides are identical. The FSR,  $n_{\text{eff}}$ , and  $n_g$  are evaluated at 1550 nm. The FSR is calculated using Eq. (5) while  $n_{\text{eff}}$  and  $n_g$  are taken from simulated data shown in Sec. III. The devices in Table II have varying width mismatches between the two waveguides, while the lengths of the two arms of the interferometer are identical. The FSR is evaluated at 1550 nm and is calculated using Eq. (9).

## III. MODELING AND SIMULATION

Waveguides with widths 450 nm to 600 nm (as listed in Tables I and II) were simulated using Lumerical MODE<sup>TM</sup> software in order to obtain mode profiles and refractive indices. The simulation results presented here are for the TE mode.

Figs. 2 and 3 display the simulated values of  $n_{\text{eff}}$  and  $n_g$  as functions of wavelength from 1500 nm to 1600 nm. Note that while  $n_{\text{eff}}$  increases with increasing waveguide width,  $n_g$  decreases.

The values extracted from these simulations allow a compact model of the form of Eq. (1) to be written for these specific waveguides. The extracted values of  $n_1$ ,  $n_2$ , and  $n_3$  are listed in Table III.

Simulated mode profiles are shown in Fig. 1. The geometries of the waveguides are shown superimposed as white

TABLE III  
WAVEGUIDE COMPACT MODEL PARAMETERS

| Width  | $n_1$ | $n_2$ ( $\mu\text{m}^{-1}$ ) | $n_3$ ( $\mu\text{m}^{-2}$ ) |
|--------|-------|------------------------------|------------------------------|
| 450 nm | 2.353 | -1.251                       | -0.037                       |
| 500 nm | 2.445 | -1.131                       | -0.043                       |
| 550 nm | 2.513 | -1.036                       | -0.027                       |
| 600 nm | 2.567 | -0.962                       | -0.008                       |

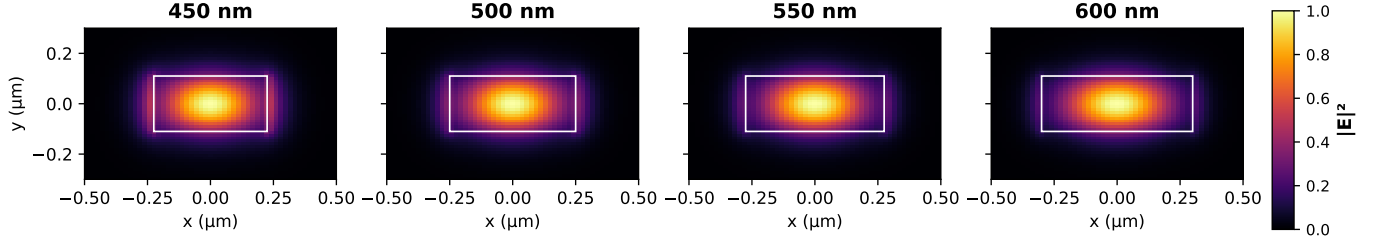


Fig. 1. Simulated mode profiles (electric field intensity  $|E|^2$ ) at varying waveguide widths.

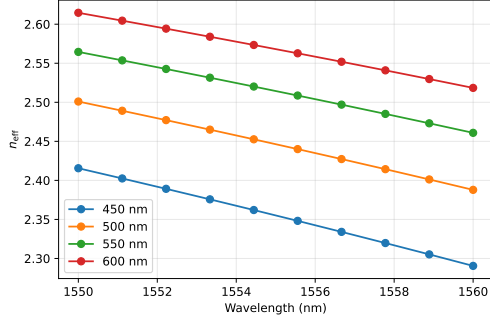


Fig. 2. Effective indices  $n_{\text{eff}}$  for waveguides of varying width.

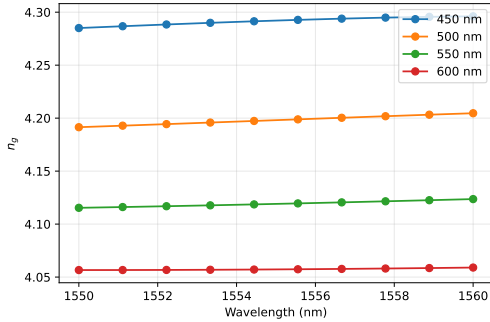


Fig. 3. Group indices  $n_g$  for waveguides of varying width.

rectangles. Notice that for the narrower waveguide (left-most), the mode profile leaks out the edges of the waveguide more so than in does for the widest waveguide (right-most). This size and strength of this evanescent wave is what determines the effective waveguide index and also has effects on the propagation loss of the waveguide.

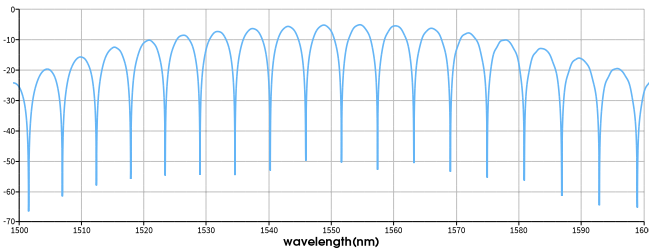


Fig. 4. Simulated transmission of device 1, a MZI with  $\Delta L$  of 100  $\mu\text{m}$  and waveguide width 500 nm

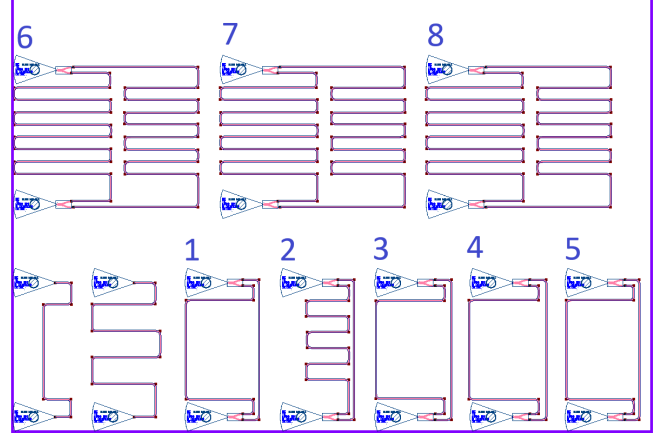


Fig. 5. Layout of the devices to be fabricated. Two de-embedding devices (not numbered) are included, consisting of only input and output grating couplers and a simple waveguide connecting them.

The optical transfer function is simulated for device 1 (see Table I) using Lumerical INTERCONNECT™. The output is shown in Fig. 4. Clear dips are visible at wavelengths where total destructive interference is occurring. The overall shape of the spectrum including the fact that it never reaches a no-loss wavelength (which would correspond to 0 dB) are due to the wavelength dependent loss of the grating couplers used to get light on and off of the silicon photonic chip.

The FSR of the simulated spectrum is approximately 5.7 nm. This agrees well with the calculated value shown in Table I. This extracted FSR value can be used to calculate the group index  $n_g$  according to Eq. (6) which results in a value of 4.21. This agrees well with the value of 4.20 that can be calculated based on the parameters of Table III with Eq. (2).

#### IV. FABRICATION

The devices listed in Tables I and II were fabricated using electron-beam lithography. An image of the layout file that was sent for manufacturing is shown in Fig. 5. Noted on the image are the eight devices listed in the tables, as well as two de-embedding devices which consist of only two grating couplers connected by a simple waveguide (of lengths 200  $\mu\text{m}$  and 400  $\mu\text{m}$ ). These devices can be used as a rudimentary test of waveguide and grating coupler losses.

TABLE IV  
ANTICIPATED VARIATION IN  $n_{\text{eff}}$  AT 1550 nm DUE TO MANUFACTURING VARIABILITY.

| Waveguide width | Min $n_{\text{eff}}$ | Nominal $n_{\text{eff}}$ | Max $n_{\text{eff}}$ |
|-----------------|----------------------|--------------------------|----------------------|
| 450 nm          | 2.27                 | 2.35                     | 2.39                 |
| 500 nm          | 2.38                 | 2.45                     | 2.47                 |
| 550 nm          | 2.46                 | 2.51                     | 2.54                 |
| 600 nm          | 2.52                 | 2.57                     | 2.59                 |

TABLE V  
ANTICIPATED VARIATION IN  $n_g$  AT 1550 nm DUE TO MANUFACTURING VARIABILITY.

| Waveguide width | Min $n_g$ | Nominal $n_g$ | Max $n_g$ |
|-----------------|-----------|---------------|-----------|
| 450 nm          | 4.27      | 4.29          | 4.36      |
| 500 nm          | 4.18      | 4.19          | 4.25      |
| 550 nm          | 4.11      | 4.12          | 4.16      |
| 600 nm          | 4.05      | 4.06          | 4.09      |

#### A. Fabrication variability and corner analysis

Electron-beam lithography can be very precise. However, there will likely still be small and unpredictable manufacturing variability in the fabricated waveguides. This must be taken into account when analyzing the experimental data and extracting parameters such as the group index.

Here we will do simple corner analysis to anticipate the manufacturing variability in waveguide height and width according to specifications provided in the course. These specifications are that (1) the nominal waveguide height is 220 nm but may be as small as 215.3 nm or as large as 223.1 nm and (2) the nominal width is 500 nm but may be as narrow as 470 nm or as wide as 510 nm. Corner analysis entails running simulations for the nominal values as well as the four permutations of combinations between the most extreme width and height values.

Several of the devices included in this report (devices 3–8) have non-standard waveguide widths. Since information on anticipated manufacturing width variability was not provided by the class for these waveguide widths, the assumption was made that they will vary by the same amount as the standard 500 nm waveguides, namely becoming narrower by up to 30 nm or wider by up to 10 nm.

Based on the corner analysis simulations, the anticipated variability in  $n_{\text{eff}}$  and  $n_g$  at 1550 nm is displayed in Tables IV and V, respectively. The nominal values of  $n_{\text{eff}}$  and  $n_g$  are included for convenient reference. For all waveguide widths and for both  $n_{\text{eff}}$  and  $n_g$ , the variations were primarily caused by the changes in waveguide width, with variability in waveguide height causing variations generally less than half of the magnitude of those caused by the waveguide width variations.

Using the predicted variations in  $n_g$  shown in Table V, we can anticipate the potential variation in the free spectral range (FSR) of the fabricated devices. This is done using Eqs. 5 and 9, and the results are shown in Table VI.

Of particular note is that huge variability is possible in the FSR of devices 6–8. These devices have identical waveguide lengths between the two interferometer arms, but varying

TABLE VI  
ANTICIPATED VARIATION IN DEVICE FSR AROUND 1550 nm DUE TO MANUFACTURING VARIABILITY.

| Device | Key Parameter | Min FSR | Nominal FSR | Max FSR |
|--------|---------------|---------|-------------|---------|
| 1      | $\Delta L$    | 5.65 nm | 5.72 nm     | 5.75 nm |
| 2      |               | 2.83 nm | 2.86 nm     | 2.87 nm |
| 3      |               | 5.52 nm | 5.60 nm     | 5.62 nm |
| 4      |               | 5.77 nm | 5.83 nm     | 5.85 nm |
| 5      |               | 5.87 nm | 5.92 nm     | 5.94 nm |
| 6      | $\Delta n_g$  | 15.1 nm | 27.9 nm     | 136 nm  |
| 7      |               | 18.3 nm | 34.1 nm     | 171 nm  |
| 8      |               | 13.0 nm | 19.0 nm     | 31.0 nm |

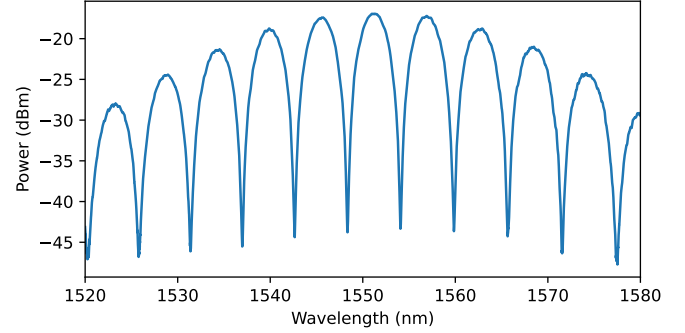


Fig. 6. Experimentally measured of device 1, a MZI with  $\Delta L$  of 100  $\mu\text{m}$  and waveguide width 500 nm

widths, leading to a variation in  $n_g$ . The key parameter in determining the FSR for these interferometers is the difference in  $n_g$  between the two waveguides. When that difference—when accounting for potential manufacturing variability in the corner analysis method used here—becomes very small, it can cause a huge increase in the predicted FSR because it sits in the denominator of Eq. 9.

This leads to the anticipation that the extracted FSR and associated  $n_g$  values taken from the experimental data for devices 6–8 may be relatively unreliable. However, the question arises of how likely this extreme variation in FSR is to occur. It is reasonable to assume that fabrication variations in waveguide width may be relatively consistent over a small area of the chip (and perhaps even over the entire chip, primarily occurring on a chip-to-chip scale). With this assumption, then—especially on the scale of a single interferometer device which takes up a small footprint—it is probable that if one of the waveguides is narrower or shorter than planned, the other waveguide will also be somewhat narrower and shorter. This correlation in fabrication error will work to avoid the extreme swings in FSR because the extremes occur for fabrication errors where, for example, one waveguide is much smaller than anticipated while the other is much larger.

#### V. EXPERIMENTAL RESULTS AND ANALYSIS

The experimentally measured transmission of device 1 is shown in Fig. 6. Easily visible is an overall curving shape to the data. This is not a feature of the interferometer structure but is instead caused by the wavelength dependent loss of the grating couplers.

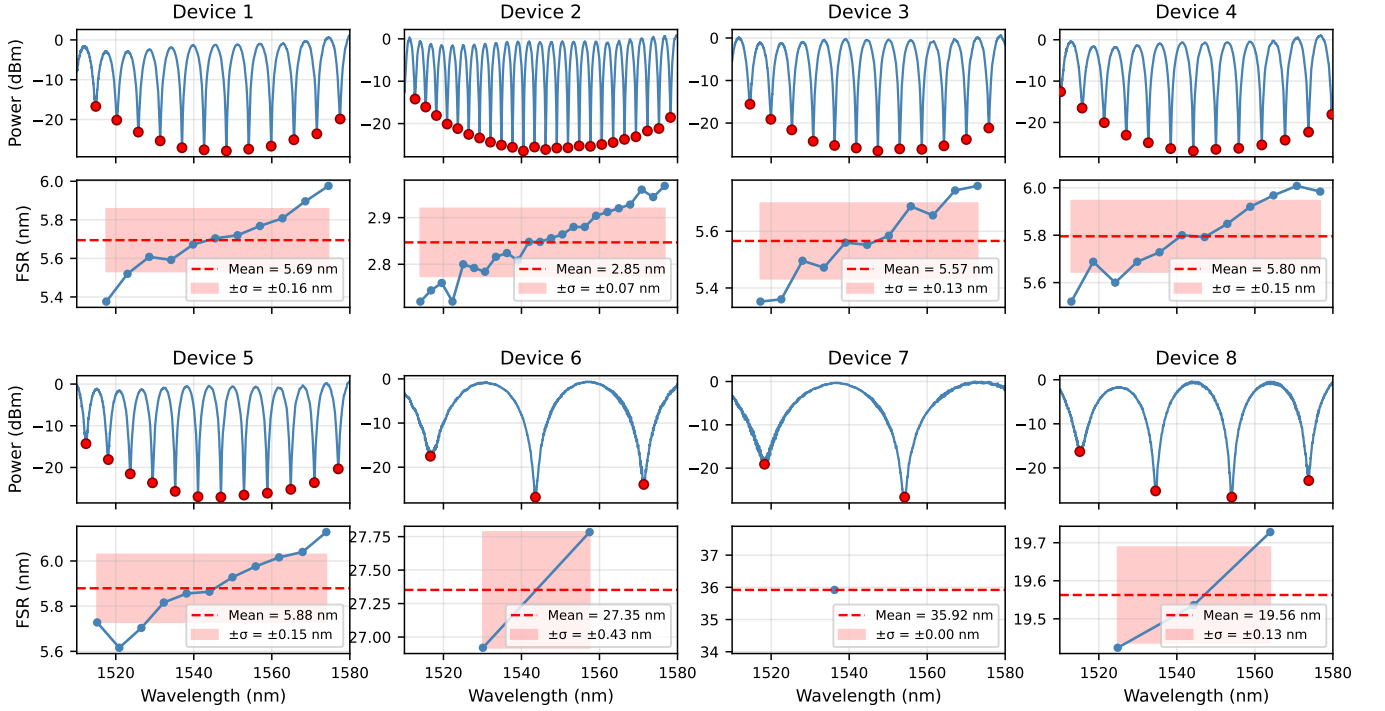


Fig. 7. Top row: illustration of peak finding. Bottom row: extracted FSR values.

As described in Sec. IV, two waveguide-only devices (without device numbers) were fabricated to help remove the effects of the grating couplers. These were fabricated with two different waveguide lengths with the intent of decoupling both grating coupler loss and linear propagation loss. Analysis of the transmitted power from these two devices showed that on average the power returned matched very closely, differing by only 0.08 dBm with a standard deviation of 0.31 dBm over the range 1510 nm to 1580 nm. Thus it was concluded that the length difference between the two waveguides was insufficient to determine the linear propagation loss with any statistical certainty. In the remaining analysis, the transmission responses are analyzed using grating-coupler-calibration based on the waveguide-only device with 200  $\mu\text{m}$  length.

The transmission response of the 200  $\mu\text{m}$  waveguide was fit using a third-order polynomial and a least-squares fitting algorithm over the same range of 1510 nm to 1580 nm. This polynomial is then subtracted from the transmission data for each of the interferometer devices.

Once a flattened spectrum is obtained, the free spectral range (FSR) of each device was obtained from the measured transmission. The process of finding the FSR is illustrated in Fig. 7. This was done by first identifying all the minima, as illustrated in the first row of the figure. These correspond to destructive interference at the second y-branch of the interferometer. The spacing between subsequent peaks is a good measure of the FSR. This is plotted in the second row of the figure. The upward trend in FSR common to all the devices is due to the fact that FSR is more accurately a constant function when found in terms of *frequency*, not wavelength, and wavelength is inversely proportional to frequency. This

TABLE VII  
PREDICTED FSR AND  $n_g$  COMPARED WITH VALUES EXTRACTED FROM EXPERIMENTAL DATA

| Device | Pred. FSR | Meas. FSR | Pred. $n_g$          | Meas. $n_g$ |
|--------|-----------|-----------|----------------------|-------------|
| 1      | 5.72 nm   | 5.69 nm   | 4.19                 | 4.22        |
| 2      | 2.86 nm   | 2.85 nm   | 4.19                 | 4.22        |
| 3      | 5.60 nm   | 5.57 nm   | 4.29                 | 4.32        |
| 4      | 5.83 nm   | 5.80 nm   | 4.12                 | 4.15        |
| 5      | 5.92 nm   | 5.88 nm   | 4.06                 | 4.09        |
| 6      | 27.9 nm   | 27.4 nm   | $\Delta n_g : 0.096$ | 0.098       |
| 7      | 34.1 nm   | 35.9 nm   | $\Delta n_g : 0.078$ | 0.074       |
| 8      | 19.0 nm   | 19.6 nm   | $\Delta n_g : 0.140$ | 0.136       |

appears as a somewhat linear trend in FSR as a function of wavelength when using the less-accurate method used here.

The calculated mean FSR for each device is shown in Table VII in the "Meas. FSR" column. Additionally shown, in the column labeled "Meas.  $n_g$ ," is the value of  $n_g$  (or  $\Delta n_g$  for devices 6–8) computed using the mean FSR and Eq. 6 (Eq. 9 for devices 6–8). The predicted theoretical values of FSR and  $n_g$  are shown for quick reference.

Comparison with Tables VI and V show that the experimentally measured FSR and  $n_g$  all fit within the expected manufacturing variability based on the corner analysis. This provides strong evidence to suggest that the devices were manufactured with good precision. For devices 1–5, the experimentally measured  $n_g$  is in every case larger than the predicted value. This is consistent with fabricated waveguides being narrower in width than the intended geometry, as noted in Sec. IV-A.

## REFERENCES

- [1] L. Chrostowski and M. Hochberg, *Silicon Photonics Design*, 1st ed. Cambridge University Press, 2015.